Amazon EFS Introduction

Explore Amazon Elastic File System, a scalable and managed file storage solution enabling multiple EC2 instances to access shared data with POSIX compliance. Learn deployment models, performance modes, throughput options, and lifecycle management to optimize cost and availability for cloud applications.

Amazon Elastic File System (EFS) is a fully managed, scalable, elastic file storage service that allows multiple Amazon EC2 instances to access a common dataset in parallel, using the standard Network File System (NFS) protocol. What sets EFS apart is that it scales automatically as files are added or removed, so there's no need to provision storage in advance.

Think of EFS as a shared network drive, but hosted in the AWS Cloud, with built-in redundancy and high availability. Whether we're hosting a web application across multiple EC2 instances or sharing configuration files between compute resources, EFS provides a central location for file storage that behaves like a traditional file system.

Unlike Amazon S3, which stores objects, EFS is a file system in the traditic sense. Files can be structured in directories, and access is POSIX-compliant, meaning standard Linux file permissions and operations apply.

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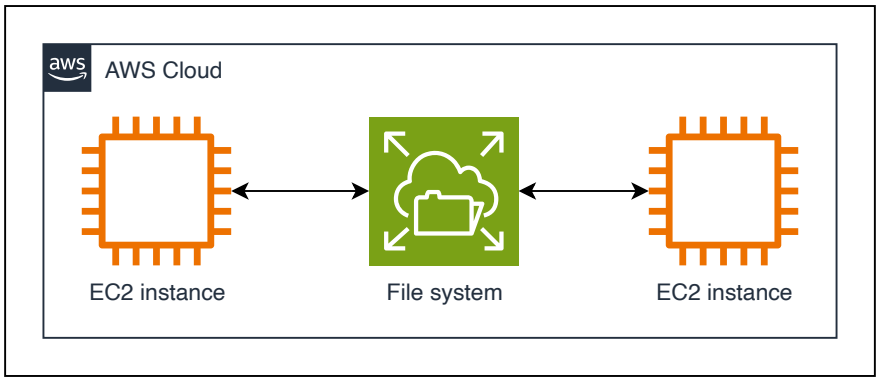

EFS attached to two EC2 instances

EFS supports both burstable and provisioned throughput modes, and we only pay for what we use. This makes it a cost-effective and operationally simple solution for workloads that need shared file access without managing hardware or file server software.

Understanding EFS file server types

While Amazon EFS is a fully managed NFS-based file system, we have a couple of important decisions to make when we create a file system, especially about its durability model and cost profile. These decisions relate to what AWS calls file server types or storage classes within EFS.

EFS offers two main types of file systems:

- **Regional file system:** Regional file system is deployed in multiple AZs across a region. Because they redundantly backup files in multiple AZs, they are highly available. It requires a mount target in each availability zone to allow the connection of resources. The illustration given below shows how different EC2s can connect to EFS using mount targets:

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Regional file system

> **Note:** In regional file systems, multiple subnets in a single AZ can connect through the same mount target.

- Here's the AWS CLI command to create a regional EFS system:

main.sh

```
aws efs create-file-system \
  --performance-mode generalPurpose \
  --throughput-mode bursting \
```

```
--tags Key=Name,Value=MyEFS
```

Create EFS using AWS CLI

- Here's a brief explanation of the command:
  - `aws efs create-file-system`**:** It initiates the creation of a new EFS file system.
  - `--performance-mode`**:** It sets the file system to use general purpose performance mode, ideal for latency-sensitive workloads like web servers.
  - `--throughput-mode`**:** It enables bursting throughput, which automatically adjusts performance based on file system size.
  - `--tags`**:** It adds a tag to the file system to identify it by name in the console or CLI.

- **One zone file system:** It is located in a single zone and can have only one mount target in the zone. This results in lower cost and reduced latency compared to regional file systems. They are designed for use cases that do not require Multi-AZ redundancy. While the file system is limited to one AZ, resources in other AZs can connect to the file system through the single mount zone, but it incurs additional charges. The illustration below shows one such example where an instance in Availability Zone 2 connects to the EFS through the mount target in Availability Zone 1.

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One-zone file system

- Here's the AWS CLI command to create a one zone EFS system:

main.sh Shell

```shell
aws efs create-file-system \
  --availability-zone-name us-west-2a \
  --performance-mode generalPurpose \
  --throughput-mode bursting \
  --tags Key=Name,Value=MyOneZoneEFS
```

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Create one zone EFS

- Here, `--availability-zone-name us-west-2a` specifies that the file system will be created in a single Availability Zone, enabling the one-zone storage class (cost-optimized).

EFS options

Amazon EFS is designed to be flexible, and part of that flexibility comes from the ability to choose how our file system behaves under load. Performance modes determine how our EFS file system manages metadata and file access under different workloads.

Elastic file systems have two performance modes:

- **General purpose (default):** This is best for most workloads, especially latency-sensitive ones like web servers, content management systems, or home directories. It offers lower latency for metadata operations.
- **Max I/O:** This mode is designed for highly parallelized workloads, like big data processing or large-scale media workflows. Due to its distributed metadata model, it offers higher throughput and lower latencies. It is useful when thousands of EC2 instances are accessing the file system concurrently.

It also offers two types of throughput modes:

- **Bursting throughput (default):** This option automatically scales throughput as our file system grows. It is ideal for unpredictable workloads or small-to-medium workloads, and it is great for short bursts of high performance.
- **Provisioned throughput:** This options lets us specify how much throughput we want, independent of storage size. It is useful for workloads with hi throughput demands but small file system sizes.

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If we're building a logging system where multiple EC2 instances write logs in parallel, max I/O mode with provisioned throughput may be more suitable. On the other hand, a web application with moderate traffic likely performs better with general purpose and bursting throughput.

Lifecycle management

Both regional and one-zone file systems can enable lifecycle management, which automatically transitions files that haven't been accessed for a set period (7, 14, 30, 60, or 90 days) to the Infrequent Access (IA) storage class.

- IA storage reduces cost by ~92% for cold data.
- Files are automatically moved back to the standard storage class upon access.
- It is transparent to the application layer.

Enabling lifecycle management cuts costs without rewriting any application logic.

Lifecycle management with EFS

Conclusion

Amazon EFS delivers a simple, reliable, and fully managed file storage solu

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scenarios where multiple EC2 instances require simultaneous access to a shared dataset with the flexibility and performance of a traditional file system. By

supporting both regional and one-zone deployment models, multiple performance and throughput modes, and integrated lifecycle management, EFS adapts seamlessly to various workloads, from highly available, Multi-AZ production systems to cost-sensitive, single-AZ environments.

Its ability to scale automatically, enforce POSIX permissions, and optimize costs through infrequent access tiers allows us to focus on the application rather than

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